Task 3: Rules, Data and Insights

Final Project - Task 3 | Digital Loan Pre-Approval & Onboarding (MVP), Online Bank | Prepared by: [Your Name], Business Analyst | 09 November 2026

Part 1: Business Rules Catalog

> **AI prompt used:** *Generate 8 business rules for an online bank's digital loan onboarding process. Include rules for application completeness, KYC verification, soft credit pulls, minimum credit score thresholds, retry logic for third-party services, and decision routing. Present each rule with ID, name, description, trigger, and IF/THEN logic with threshold values.*

Rule ID	Rule name	Description	Trigger condition	Logic / expected behavior
BR-01	Application completeness	All mandatory fields (name, DOB, address, SSN/ID, income, loan amount, term) must be present and valid before processing.	Application submitted	IF any mandatory field is missing or invalid THEN status = "Incomplete" and highlight the fields; no KYC or credit call is made
BR-02	Consent before credit pull	A soft credit pull requires recorded, unexpired consent.	Before credit pull	IF soft-pull consent is missing or older than 30 days THEN block the pull and request consent again
BR-03	KYC verification outcome	Identity must be verified before any approval.	KYC result received	IF match = full AND no watchlist hit THEN status = "Verified"; IF partial match THEN route to manual review; IF no match OR watchlist hit THEN status = "KYC failed" and stop
BR-04	Eligibility basics	Applicant and loan must be within product limits.	Application submitted	IF age < 18 OR amount outside $1,000-$50,000 OR term outside 12-60 months THEN decline with reason code "Not eligible"
BR-05	Credit thresholds	Soft-pull results drive the automated decision.	Credit score received	IF score >= 680 AND DTI <= 40% THEN "Pre-approved"; IF 600 <= score < 680 OR 40% < DTI <= 50% THEN manual review; IF score < 600 OR DTI > 50% THEN "Declined"
BR-06	Retry and backoff for third-party APIs	Transient failures must not decline customers.	KYC or bureau call fails / times out (> 10 s)	IF call fails THEN retry up to 3 times with exponential backoff (2 s, 4 s, 8 s); IF still failing THEN status = "Pending - service delay", route to manual review and notify the customer
BR-07	No credit file (thin file)	Applicants without a bureau file need human review.	Bureau returns no-hit	IF bureau response = no record THEN route to manual review (never auto-decline)
BR-08	Decision routing and SLA	Manual cases must be decided on time.	Case enters review queue	IF case age > 4 business hours THEN escalate to the Operations supervisor
BR-09	Adverse action notice	Declined applicants must be told the main reasons.	Decision = Declined	IF declined THEN send an adverse-action notice with up to 4 reason codes and the credit bureau details
BR-10	Fraud / velocity flag (optional)	Detect suspicious repeat applications.	Application submitted	IF > 3 applications from the same device or SSN within 24 hours THEN flag "Risk review" and route to manual review

Review notes: BR-02 (consent), BR-07 (no credit file) and BR-09 (adverse action) cover gaps raised in the Task 1 interviews. BR-06 routes API failures to review instead of declining, because a system error must not decline a customer. Thresholds are proposals to confirm with Risk/Credit.

Part 2: Data Entities and Attributes

> **AI prompt used:** *Create a table of core data entities used in a digital loan onboarding system for an online bank. Include entities Customer, Application, CreditPull, KYCCheck, Consent and AuditLog. For each entity, provide 6-8 attributes with field types and descriptions, and state the relationships.*

Entity	Attributes (field: type)	Description
Customer	customer_id (PK, UUID); full_name (VARCHAR 100); date_of_birth (DATE); email (VARCHAR 120); phone (VARCHAR 20); residential_address (VARCHAR 250); ssn_token (VARCHAR 64, tokenized); created_at (TIMESTAMP)	The applicant (PII; encrypted at rest)
Application	application_id (PK); customer_id (FK); product_code (VARCHAR 10); loan_amount (DECIMAL 12,2); term_months (INT); annual_income (DECIMAL 12,2); channel (ENUM web/mobile); status (ENUM); decision (ENUM approve/decline/review); reason_codes (VARCHAR 100); rule_version (VARCHAR 20); submitted_at, decided_at (TIMESTAMP)	The loan request and its decision
Consent	consent_id (PK); application_id (FK); consent_type (ENUM soft_pull/esign/privacy); disclosure_version (VARCHAR 20); granted_at (TIMESTAMP); ip_address / device_id (VARCHAR 64); status (ENUM granted/withdrawn)	Evidence of customer consent
KYCCheck	kyc_check_id (PK); application_id (FK); provider (VARCHAR 50); provider_ref (VARCHAR 64); result (ENUM pass/review/fail); match_score (DECIMAL 5,2); watchlist_hit (BOOLEAN); checked_at (TIMESTAMP)	Identity verification result
CreditPull	credit_pull_id (PK); application_id (FK); pull_type (ENUM soft/hard); bureau (VARCHAR 30); credit_score (INT); dti_ratio (DECIMAL 5,2); response_status (ENUM ok/timeout/no_hit); retry_count (INT); pulled_at (TIMESTAMP)	Soft credit check result
AuditLog	audit_id (PK); application_id (FK); actor (VARCHAR 50 system/user ID); action (VARCHAR 50); details_hash (VARCHAR 128); timestamp (TIMESTAMP)	Append-only regulatory trail

Relationships

- **Customer submits Application** (1 : N)
- **Application records Consent** (1 : N, one per consent type)
- **Application triggers KYCCheck** (1 : N, re-checks allowed); **KYCCheck stores results**
- **Application requests CreditPull** (1 : N, retries stored as attempts)
- **Application is logged by AuditLog** (1 : N)

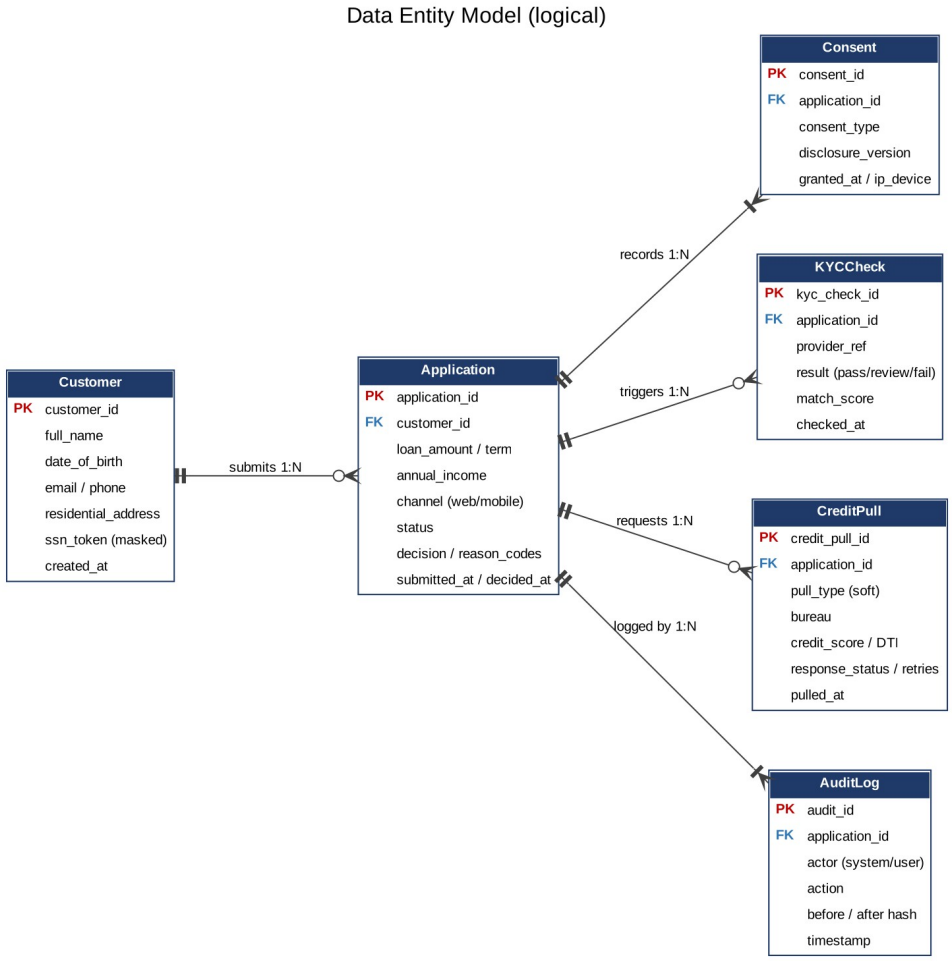


Figure 1. Logical entity model (crow's foot notation).

Part 3: Insights and Dashboard Plan

> **AI prompt used:** *Generate 8 KPIs for a loan onboarding dashboard. Include calculations for approval rate, conversion rate, average processing time, dropout rate, and KYC failure rate. Explain each KPI and suggest a target.*

KPI	Calculation	What it tells us	Target
Application completion rate	Submitted applications / started applications x 100	Form friction and drop-off	Target >= 60%
Conversion (pre-approval) rate	Pre-approved / submitted x 100	How many applicants receive an offer	Track vs credit policy
Approval rate	(Auto + manual approvals) / decided applications x 100	Overall credit outcome	Weekly trend
Average processing time	Avg(decided_at - submitted_at) for auto-decisions	Speed of the automated path	< 3 minutes
Dropout rate by stage	Applications abandoned at stage / applications reaching stage x 100	Where customers leave	Lower month on month
KYC failure rate	KYC failed / KYC checks performed x 100	Identity issues or vendor quality	< 8%
Manual review rate & SLA	Cases sent to review / decided x 100; % reviewed within 4 hours	Operations load and service	< 15%; SLA >= 95%
Third-party error rate	Failed or timed-out API calls / total calls x 100	Vendor reliability (retry logic)	< 1%

Dashboard layout (Power BI)

- **Top row:** KPI summary tiles (applications started, completion rate, approval rate, average time to decision, KYC failure rate, manual review rate)
- **Left:** funnel chart for application progression (started > consent > KYC submitted > KYC passed > credit pulled > decision)
- **Centre:** donut chart for decision outcomes (auto approved / auto declined / manual review)
- **Right:** bar chart of KYC failure reasons
- **Bottom:** trend line for weekly approval rate against target
- **Filters:** date range, channel (web/mobile), product, decision path; data refreshed hourly from the Application, KYCCheck and CreditPull tables

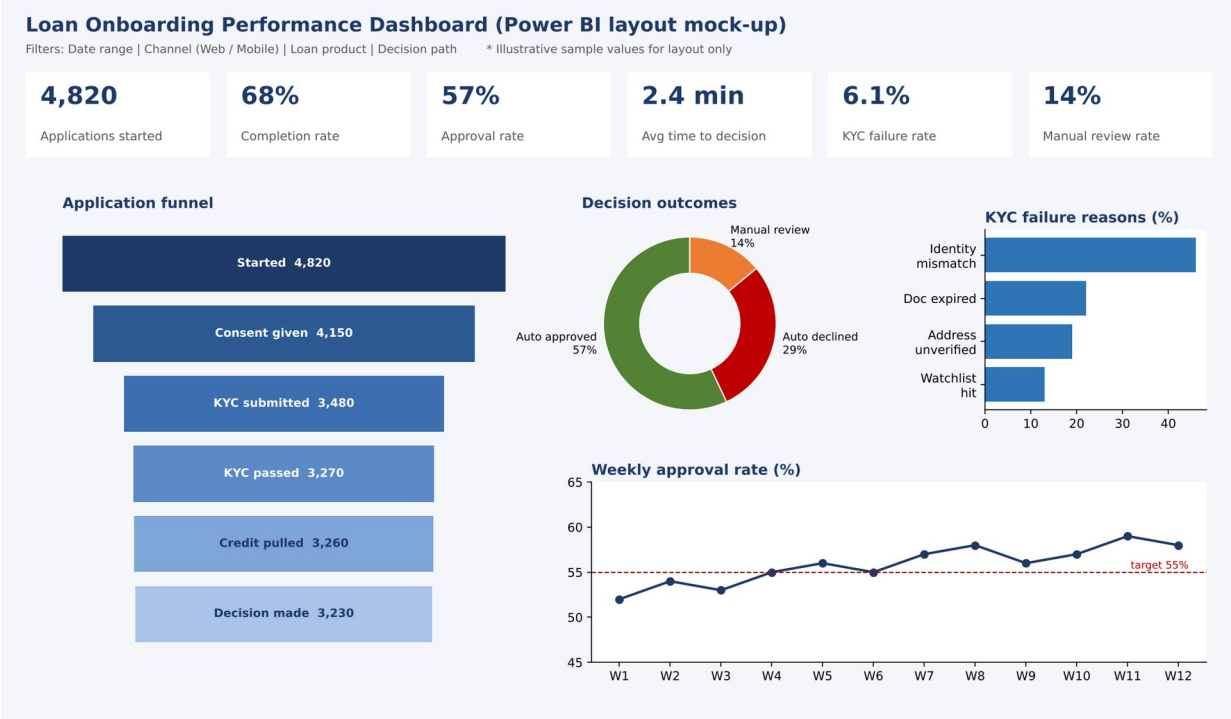


Figure 2. Dashboard layout mock-up (values are illustrative placeholders).